David Duong

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Education

University of California, Irvine — Irvine, CA
Bachelor of Science in Computer Science - 3.92/4.00 — *December 2027*

Experience

Ticketmaster — Oct. 2026 - Present
Software Engineer Intern — *Beverly Hills, CA*

- Interning on the Cassandra infrastructure team.

First American — Jun. 2026 - Aug. 2026
Data Engineer Intern — *Santa Ana, CA*

- Developed a data validation dashboard with React, FastAPI, and PostgreSQL to enable automated comparison of tables from Databricks, BigQuery, and SQL Server to support data migration efforts.
- Built and benchmarked cross-platform data validation services using Arrow, Parquet, Spark, DuckDB, and Declarative Automation Bundles to perform row-level comparisons, processing 100M+ row datasets in 5 minutes.
- Architected a persisted job queue in PostgreSQL to orchestrate concurrent validation jobs to ensure reliability.

Temco Logistics — Mar. 2026 – Jun. 2026
Data Engineer Intern — *Carson, CA*

- Engineered a fuzzy-matching based data pipeline using Python and Spark in Databricks to migrate 50,000+ records from a legacy platform to cloud storage to serve internal legal, operations, and HR teams.
- Refactored a legacy ingestion service from pandas to Spark, reducing the codebase by over 50% with Azure Key Vault and Blob Storage integration.
- Optimized Spark transformation runtimes from over 2 hours to 30 minutes by utilizing multi-threading and partitioning strategies to maximize throughput.

CareTech UCI — Oct. 2025 – Apr. 2026
Software Developer — *Irvine, CA*

- Developed a full-stack application with a team of 8 developers using React, FastAPI, and Supabase to track and visualize daily vitamin intake to promote cognitive health.
- Trained a YOLO26 computer vision model for meal classification by preprocessing image data through resizing, normalization, and augmentation, and iterating across model sizes to achieve 80%+ classification accuracy.

Projects

Sentiment Data Pipeline | *Go, Python, SQL*

- Engineered services using Go and Python to ingest and classify 1,000+ data points per second.
- Optimized data ingestion by over 80% through the implementation of concurrent worker pools in Go.
- Implemented classification for 5,000+ datapoints by integrating a Hugging Face transformers model.

NASA Kibo Robot Programming Challenge | *Java, Python, PyTorch*

- Ranked 6th out of 55 teams nationally by engineering an Android application that autonomously commanded NASA's Astrobee robot to capture and classify images using NASA's APIs and OpenCV.
- Engineered a multithreaded Java pipeline to parallelize image capture and robot movement, maximizing efficiency.
- Built a data augmentation pipeline with Python, PyTorch, and Albumentations to synthetically generate training data to achieve 85%+ accuracy on a YOLOv5 model.

Technical Skills

Languages: Python, Go, SQL, Typescript, Java, Bash
Technologies: Spark, Arrow, PostgreSQL, Delta Lake, Parquet, DuckDB, SQL Server, PyTorch
Frameworks: FastAPI, Node.js/Express.js, React
Cloud Platforms: Databricks (Lakebase, Lakeflow, Foundation Models), GCP (Cloud Storage, BigQuery, Agent Platform), Azure (Blob Storage, SQL Server, Key Vault)
Developer Tools: Git, Github Actions, Docker, Linux, Cursor, Codex